

Mathematical modelling: Homework 2

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Initial reasoning

We have 2 curves K and L in $\mathbb{R}^2$. We want to find all the intersection points of K and L . K is parametrized by $p(t)$ and is defined on an interval $I = [a, b]$. L is parametrized by $q(t)$ and is defined on an interval $J = [c, d]$. We want to find the points of intersection using the following procedure:

1. Divide I and J into subintervals of length $h > 0$. This means we will have subintervals:

$$[a, a+h], [a+h, a+2h], \dots, [a+kh, b]; \quad k = \text{floor}\left(\frac{b-a}{h}\right)$$

and

$$[c, c+h], [c+h, c+2h], \dots, [c+ih, d]; \quad i = \text{floor}\left(\frac{d-c}{h}\right)$$

. The edges of the intervals will be exactly the points that we later use to approximate K and L with polygonal chains.

2. Approximate K and L with polygonal chains determined by evaluating the parametrizations at subdivision points of I and J and find the intersections of the polygonal chains.

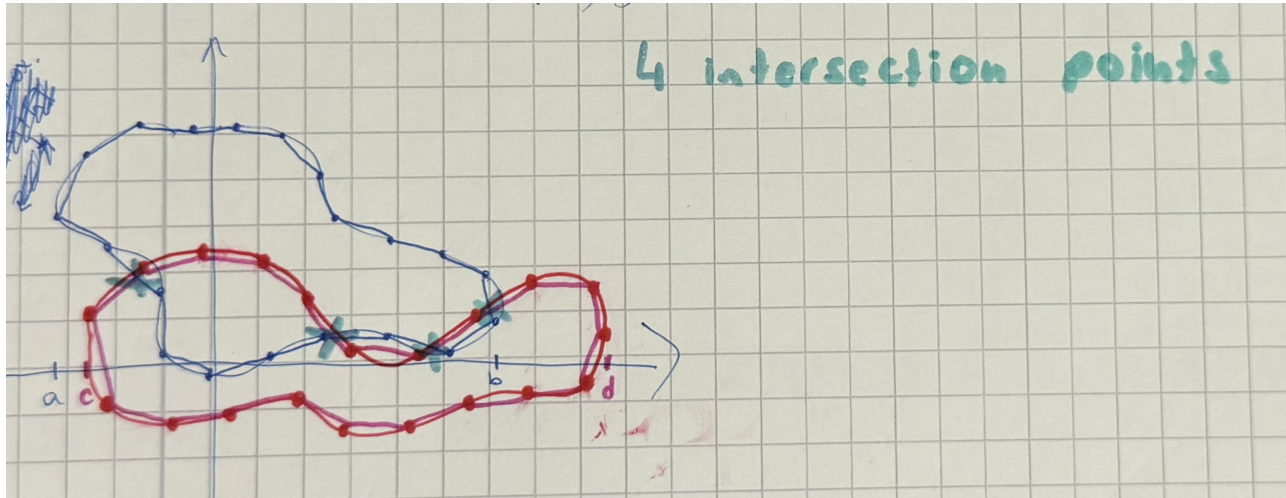


Figure 1: How I imagine the approximation and its intersections will look

3. Use the points of intersection of the polygonal approximations as an initial guess for Newton's iteration, which will determine the actual intersection points of K and L more accurately.

Task 1

To find all intersection points of curves parametrized by p and q means one needs to find all the solutions of the equation $p(t) = q(u)$. Suppose that $P'(x_i, y_i)$ is one of the intersection points of the polygonal chains K' and L' . What is a sensible choice for the values of parameters $t_i^{(0)}$ and $u_i^{(0)}$, such that $x_i^{(0)} = [t_i^{(0)} \ u_i^{(0)}]^T$ is a good initial approximation for Newton's iteration? (That is: an $x_i^{(0)}$ which converges to $x_i = [t_i \ u_i]$, such that $p(t_i) = q(u_i)$ is precisely the spatial vector of the intersection point P , closest to P' .)

Solution

First let's look at what we even need to find. We are given a point in the $\mathbb{R}^2$ plane with coordinates (x_i, y_i) which is one of the intersection points of the polygonal chains K' and L' . For starters let's say our K' and L' are made out of line segments connecting pairs of points $p(t_i), p(t_{i+1})$ where $i = 1, \dots, n-1$, and $q(u_j), q(u_{j+1})$ where $j = 1, \dots, m-1$ and the i -th line segment of K' intersects with the j -th line segment of L' at our intersection point (x_i, y_i) . We will specify later what n and m are.

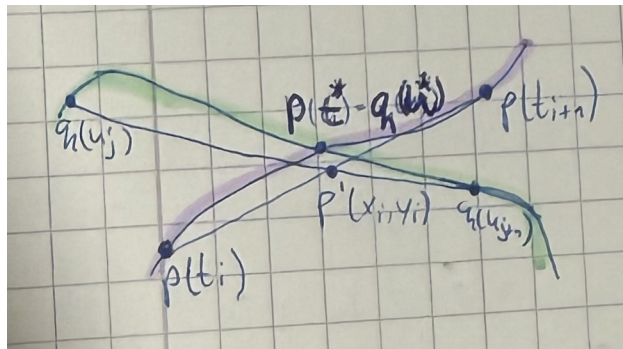


Figure 2: The true intersection point $p(t^*) = q(u^*)$ of the curves and the approximated intersection point of polygonal chains $P'(x_i, y_i)$

We now need to approximate t^* and u^* well enough for Newton's iteration to converge. If we parametrize the line segment between $p(t_i)$ and $p(t_{i+1})$ we get:

$$p_i(s) = p(t_i) + s \cdot (p(t_{i+1}) - p(t_i)) \quad s \in [0, 1]$$

and the line segment between $q(u_j)$ and $q(u_{j+1})$

$$q_j(r) = q(u_j) + r \cdot (q(u_{j+1}) - q(u_j)) \quad r \in [0, 1]$$

Now let's set the value of $p_i(s)$ and $q_j(r)$ as the point (x_i, y_i) , after all we know that they intersect there. $p(t_i), p(t_{i+1}), q(u_j), q(u_{j+1})$ are all known. To find how far along the line segment we went we can use the formula:

$$s = \frac{\|(x_i, y_i) - p(t_i)\|}{\|p(t_{i+1}) - p(t_i)\|}$$

and

$$r = \frac{\|(x_i, y_i) - q(u_j)\|}{\|q(u_{j+1}) - q(u_j)\|}$$

After acquiring s and r we can approximate t^* and u^* as:

$$t^* = t_i + s \cdot (t_{i+1} - t_i)$$

and

$$u^* = u_j + r \cdot (u_{j+1} - u_j)$$

The local parameter $s \in [0, 1]$ describes the relative position of the point (x_i, y_i) along the segment between $p(t_i)$ and $p(t_{i+1})$. We can map this local parameter back to the global parameter interval $[t_i, t_{i+1}]$ with the affine transformation above. This is a good approximation if our h is sufficiently small because then locally the curve will appear "almost straight" if we aren't unlucky and the curve has a massive spike there or something. An analogous argument holds for u^* . Therefore $t_i^{(0)} = t^*$ and $u_i^{(0)} = u^*$.

Task 2

Write the system of equations $p(t) = q(u)$ in a form suitable for Newton's iteration and express the corresponding Jacobian matrix using derivatives $\dot{p}$ and $\dot{q}$.

Solution

Let's first define $p(t) = [f_1(t) \ f_2(t)]^T$ and $q(u) = [g_1(u) \ g_2(u)]^T$. We are looking for solutions of $p(t) = q(u)$ where the curves intersect. A more suitable form is $p(t) - q(u) = 0$. Now we can define $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$:

$$F(t, u) = p(t) - q(u) = \begin{bmatrix} f_1(t) - g_1(u) \\ f_2(t) - g_2(u) \end{bmatrix}$$

We need to find zeros of $F(t, u)$. The Jacobian $J_F : \mathbb{R}^2 \rightarrow \mathbb{R}^{2 \times 2}$ consists of the derivatives of the parametrizations with respect to their respective parameters, with a negative sign for the second curve due to the subtraction in $F(t, u)$:

$$J_F(t, u) = \begin{bmatrix} \frac{\partial f_1}{\partial t} & -\frac{\partial g_1}{\partial u} \\ \frac{\partial f_2}{\partial t} & -\frac{\partial g_2}{\partial u} \end{bmatrix} = [\dot{p}(t) \quad -\dot{q}(u)]$$

With $x^{(k)} = [t^{(k)} \ u^{(k)}]^T$ we can then run the Newton's method as:

$$x^{(k+1)} = x^{(k)} - J_F(x^{(k)})^{-1} \cdot F(x^{(k)})$$

or in a more standard form:

$$J_F(x^{(k)}) \Delta x^{(k)} = -F(x^{(k)}), \quad x^{(k+1)} = x^{(k)} + \Delta x^{(k)}$$

Task 3

Write a Julia function which, for curves K and L , finds all their intersection points using the above method.

Solution

Steps:

1. We calculate the number of parameters n for the polygonal chain K' and m for L' by dividing the length of the intervals with the step size h . We get the parameters using the given intervals, h and the number of them, potentially adding the last parameter separately if the length isn't divisible with h .
2. The next step is building the chains. I chose to do it by representing each polygonal chain as a list of points in $\mathbb{R}^2$ by evaluating $p(t)$ and $q(u)$ respectively with the previously calculated parameters. Here it's convenient to also remember which parameters were used to calculate which point, so in the end each element is of the following form $(pt_1, pt_2, param \text{ for } pt_1, param \text{ for } pt_2)$.
3. Calculating the intersections of the polygonal chains isn't as trivial as it may seem to us at first when we draw 2 simple polygons and look at where they cross each other.

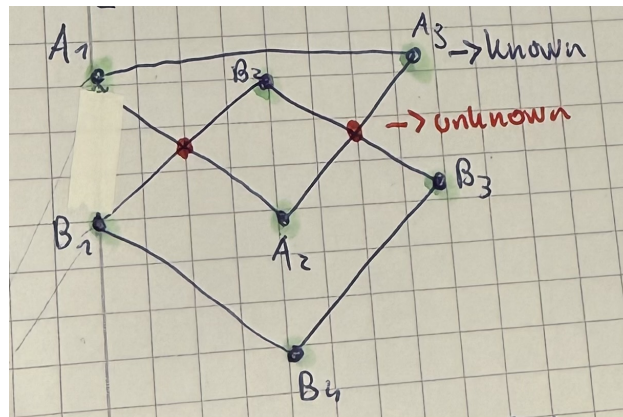


Figure 3: Simple image of 2 polygons and their intersections so we can imagine what we need to find. Green colouring shows known elements and red unknown.

The key here is to remember some Linear Algebra (or Algorithms and Data Structures 1). We can determine whether 2 line segments intersect each other by looking at relative positions of their endpoints. First let's remember that the absolute value of the determinant in the $\mathbb{R}^2$ world tells us the volume of the parallelogram spanned by 2 vectors. If we take for example points B_1, B_2, A_1 from the picture above and we define

$$u = B_2 - B_1, \quad v = A_1 - B_1,$$

then the determinant

$$d_1 = \det(u, v)$$

encodes the relative orientation of the vectors. If $d_1 > 0$, then the vector v lies to the left of u , meaning that the point A_1 lies to the left of the oriented line segment B_1B_2 . If $d_1 < 0$, it lies to the right, and if $d_1 = 0$, the points are collinear. Now we do the same for A_2 relative to B_1B_2 , we get d_2 and the other way around as well for B_1, B_2 relative to A_1A_2 and get d_3, d_4 . If we get that $d_1 \cdot d_2 < 0$ we know A_1 and A_2 lie on opposite sides of B_1B_2 . Same when we switch the roles of A_i and B_i , if we get $d_3 \cdot d_4 < 0$ we know B_1 and B_2 lie on opposite sides of A_1A_2 . It is imperative to check all 4 because we are operating with line segments and we have to guarantee the intersection is inside the segments.

If we had lines, checking one determinant would be enough. It is also possible, although unlikely, we will have an intersection such that one of the endpoints of a line segment will lie on the line segment of the other. In this case one of the determinants will be zero and we get given the point where they intersect. In the usual case of intersection though, we have to still find the point where they intersect. First we have to parametrize the line segments:

$$b(t) = B_1 + t \cdot (B_2 - B_1)$$

and

$$a(u) = A_1 + u \cdot (A_2 - A_1)$$

Then we need to solve the system of equations: $b(t) = a(u)$. We can rewrite this as:

$$B_1 + t \cdot (B_2 - B_1) = A_1 + u \cdot (A_2 - A_1) \quad (1)$$

$$t \cdot (B_2 - B_1) - u \cdot (A_2 - A_1) = A_1 - B_1 \quad (2)$$

$$t \cdot v_1 - u \cdot v_2 = b \quad (3)$$

From (2) $\rightarrow$ (3) we replace $(B_2 - B_1)$ with v_1 , $(A_2 - A_1)$ with v_2 , $A_1 - B_1$ with b . If we now define $A = \begin{bmatrix} v_1 & -v_2 \end{bmatrix}$ and $x = \begin{bmatrix} t \\ u \end{bmatrix}$ we have a system $Ax = b$, which we solve as: $x = A \backslash b$. What's left is to just input one of the parameters t or u into $b(t)$ or $a(u)$ to get the desired intersection point. Now we do this for every pair of points p_i, p_{i+1} and q_j, q_{j+1} on the polygon chains and save the results. These will serve as the initial guesses for Newton's method.

4. We run the Newton's method on every calculated polychains intersection to improve it's accuracy and save the points.
5. We return a list of improved intersection points and a list of intersection points we got from the polychains.

Task 4

Pick two "interesting" curves (with at least 3 intersection points) and include two pictures in your report: a drawing of both polygonal chains K' and L' with clearly marked intersection points, and an analogous drawing for the curves K and L .

Solution

I went ahead and googled some interesting curves and landed on a very nice looking combination (in my opinion). The first curve is the so called *love curve* and it's parametrized as:

$$p(t) = [16\sin^3(t) \quad 13\cos(t) - 5\cos(2t) - 2\cos(3t) - \cos(4t)]^T$$

It looks like a heart and its symmetric around $x = 0$. The second was the *catastrophic sine*, which is a normal sinus whose amplitude grows as $|x|$ grows. Its parametrized as:

$$q(t) = [t \quad (t + a)\sin(t)]^T \quad a \in \mathbb{R}$$

I chose $a = 0$ because that way its symmetric around $x = 0$ like the first curve. Now one thing might stand out, I specifically used a "worse" h for the polychain approximation so that the difference can be visible. Otherwise the polychain approximation already gives pretty good intersections and its not as obvious where it differs from the Newton's method improvement. Also it is clear that in the red, *love curve* polychain approximation, the number of points is a lot lower. This is because in order to nicely represent the 2 curves on one graph I had to limit the interval of the red curve to $[-2\pi, 2\pi]$ while the green was on the interval $[-20, 20]$ so with the same h we will have a lot more points for the green one. Nevertheless, it came out good in the end.

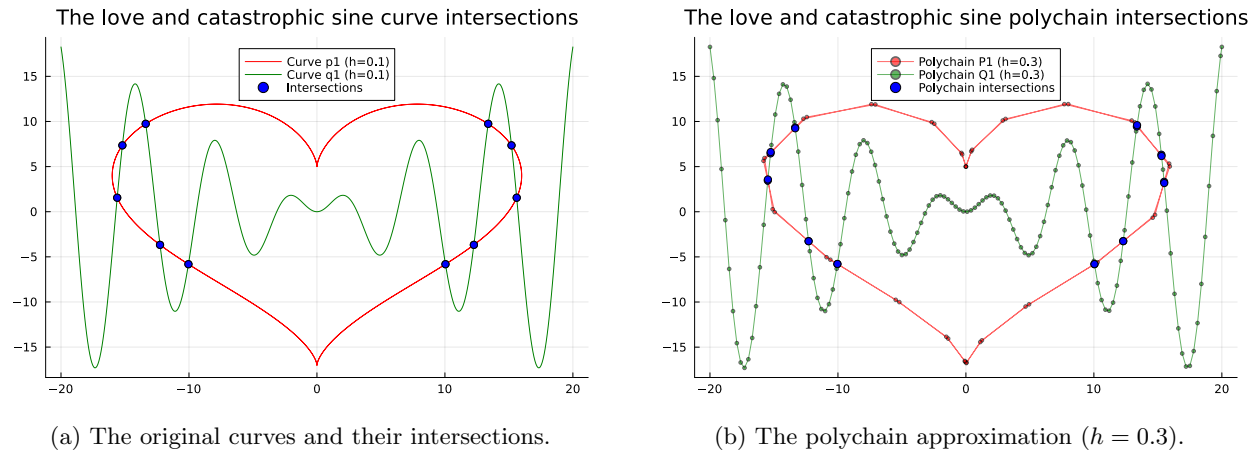


Figure 4: Comparing the 2 methods and the intersections they find.

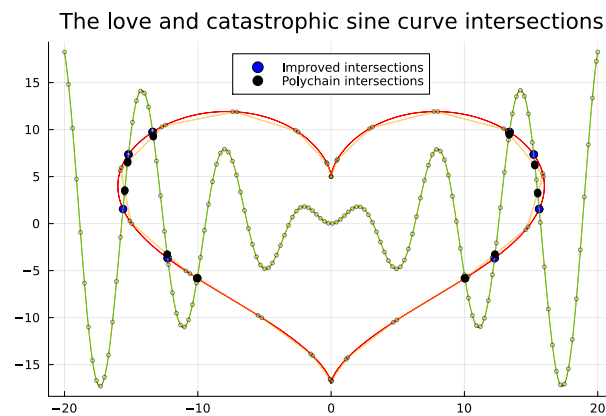


Figure 5: Both original curves and their polychain approximations together, comparing the intersections.

Conclusion

- *Who or what (AI tool) did you consult while solving this home assignment?* I used **chatgpt** and **gemini** for helping me write, correctly format and generate certain templates for this latex document.
- *Did you use AI tools for coding? To what extent is the code your own work and to what extent is it AI-generated?* I used it for error explanation and explanation of Plots how to draw nicer images.